Profiling Vision-Language Models

Performance, energy, and where the time actually goes

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Overview

1. Goal and setup
2. Baseline results
3. What didn't work
4. Where the time goes
5. Insights and next steps

The question

Project goal: understand how VLMs scale and where to spend optimization effort.

Two concrete questions:

1. Which models are Pareto-optimal on quality / latency / energy?
2. Which parts of the model dominate cost?

Approach: vary one axis at a time, measure four things directly, then split the dominant model into sub-components.

Setup

7 models, 3 datasets, 5 axes, 4 metrics

Models (7)

- `blip2-opt-2.7b`, `blip2-flan-t5-xl`
- `instructblip-flan-t5-xl`
- `instructblip-vicuna-7b`
- `llava-1.5-7b`, `llava-1.5-13b`
- `fuyu-8b`

Datasets (300 samples each):
ScienceQA, TextVQA, COCO Captions

Axes varied

- Resolution: 224 / 336 / 448 / 512
- Prompt: 10 / 50 / 100 / 200 tokens
- Batch: 1 / 2 / 4 / 8
- FP32, FP16, `torch.compile`, FA2
- GPU vs. CPU

Metrics: latency, FLOPs, energy (J), WER.
Hardware: 2× RTX 5880 Ada, 48 GiB.

What the data looks like

One sample from each dataset

ScienceQA

Q: Which animal's mouth is also adapted for bottom feeding?

A: *armored catfish*

TextVQA

Q: What number is on the player's jersey?

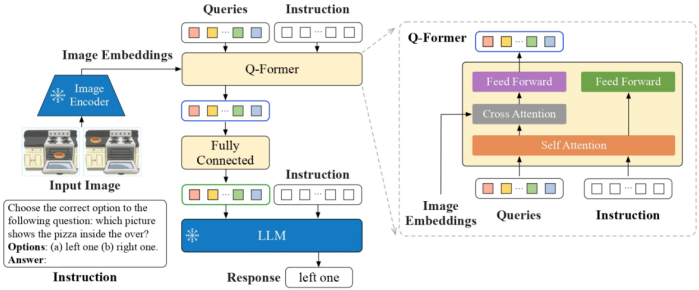
A: *22*

COCO Captions

Task: describe the image.

Ref: *A bicycle replica with a clock as the front wheel.*

Architecture: BLIP-2 / InstructBLIP



EVA ViT-g/14 (frozen, ∼1 B)

↓

Q-Former (105 M trainable) cross-attention compresses image features into 32 query tokens

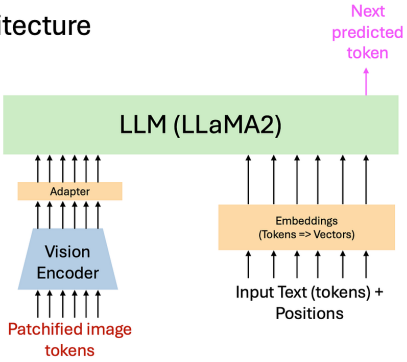
↓

LLM (frozen, OPT-2.7B or Flan-T5-XL)

InstructBLIP adds the instruction text as extra context to the Q-Former.

Architecture: LLaVA 1.5

LLaVA 1.5 Architecture



CLIP ViT-L/14 (304 M)
produces 576 patch tokens for a 336×336 image

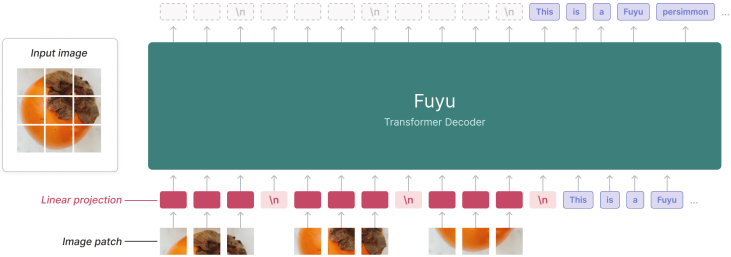
↓

MLP adapter (∼20 M)
2-layer projection into the LLM hidden size

↓

LLaMA / Vicuna (7B or 13B)
Image and text tokens are concatenated and fed into the LLM as one sequence.

Architecture: Fuyu-8B

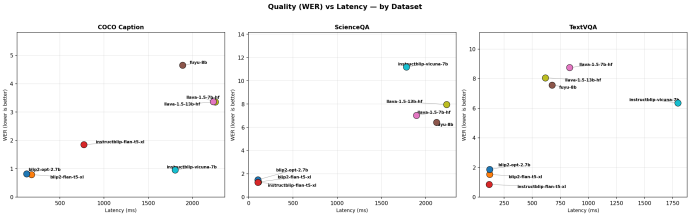


No separate vision tower.
Image is split into 30×30 patches; each patch goes through a single linear projection into the LLM hidden size.

Patch tokens are then mixed inline with text tokens and fed to a standard transformer decoder (Persimmon-8B).

Consequence: every image patch joins the KV cache and grows it on every decode step. Why per-token cost is 41 ms despite only 8 B params.

Baseline: two clear tiers



Model	Lat ms	Energy J
blip2-flan-t5-xl	136	31
blip2-opt-2.7b	115	29
instructblip-t5-xl	330	70
fuyu-8b	1 562	404
llava-1.5-7b	1 654	450
llava-1.5-13b	1 704	470
instructblip-vicuna-7b	1 793	487

Pareto winner

`blip2-flan-t5-xl`
WER 1.21, 136 ms, 31 J

7B+ tier: ∼13× slower, ∼13× more energy.

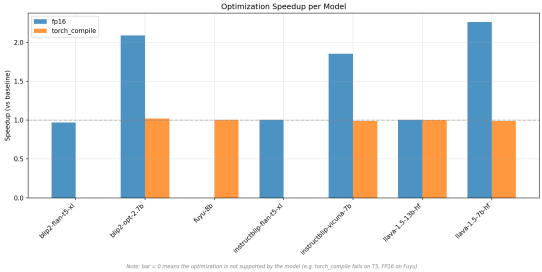
Other axes: short version

Resolution 224 → 512: flat everywhere except Fuyu ($+20\%$ at 512 — raw image patches as text tokens).

Prompt 10 → 200: T5 slows $2.3{-}3.5\times$; 7B+ decoder-only $\leq 16\%$.

Batch 1 → 8: all models gain throughput, but sub-linearly. Best: instruct-T5 / vicuna-7b at $\sim 4\times$. Worst: llava-7b at $\sim 2\times$.

CPU vs GPU: 19–30$\times$ gap. Impractical for 7B+.

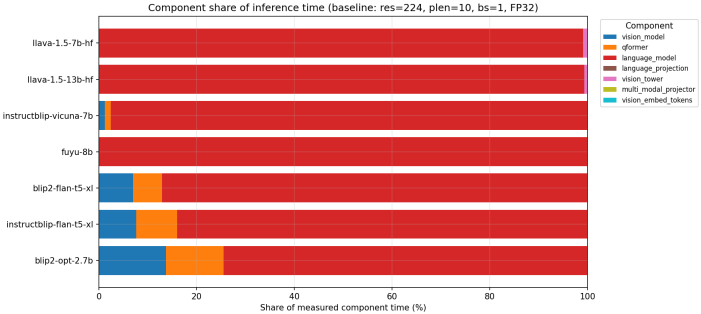


FP16 helps OPT/Vicuna/LLaMA (1.85–2.2$\times$). *Slows* T5. `torch.compile`: no measurable speedup.

What didn't work, and why

Attempted	Why it failed / what we did
`idefics2-8b`	Pixel-values shape mismatch on `transformers` 5.x ("expected 5, got 4"). Dropped from results.
`cogagent`, `moondream2`, `llava-400m-fp16`	Bespoke `trust_remote_code`; last one isn't on HF (likely typo in spec). Used `llava-1.5-7b` for same-family comparison.
`torch.compile` on T5	Compiler bails on encoder-decoder graph. No measurable speedup on the others either.
FP16 on T5	*Slower* than FP32 — internal dtype casts in T5's forward.
FP16 on `fuyu-8b`	Float / Half dtype mismatch in patch embedding.
`flash-attn` pip	Build needs `nvcc` $\geq$ 11.7; system has 11.5. Used PyTorch SDPA, which dispatches FA2 in FP16 on Ada.
Batched `fuyu-8b`	Processor returns lists for `batch>1`. Skipped batch sweep for Fuyu.

The punchline: where the time goes



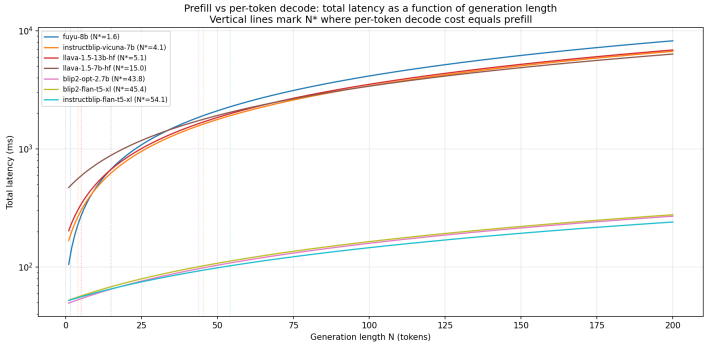
LLM decoder share

- BLIP-2 OPT 2.7b: **74%**
- BLIP-2 / Instruct-T5: 84–87%
- vicuna-7b: 98%, llava: 99%
- fuyu-8b: **99.99%**

Forward hooks + CUDA events on top-level submodules.

Optimize the decoder.

Inside the decoder: prefill vs. per-token decode



$t(N) = t_{\text{prefill}} + N\, t_{\text{decode}}$
fit at N=$1, 50$.

Crossover N^*:

- fuyu-8b: 1.6
- vicuna-7b: 4.1
- llava-13b: 5.1
- llava-7b: 15
- BLIP-2: 44–54

Short $\rightarrow$ prefill (FA2). Long $\rightarrow$ decode (spec., KV quant).

FlashAttention-2: smaller win than expected

	SDPA prefill	eager prefill	FA2 speedup	decode/tok (SDPA / eager)
`llava-1.5-7b`	94.5 ms	105.4 ms	**1.11×**	15.3 / 15.6 ms
`llava-1.5-13b`	169.0 ms	181.3 ms	**1.07×**	33.0 / 34.2 ms

Why so modest?

At plen=10 the LLaVA input is ≈ 586 tokens (10 text + 576 image patch). The eager $O(N^2)$ attention matrix still fits in cache at $N \approx 600$.

The real lever was the dtype flip

FP32→FP16 on `llava-7b`: 4.6× prefill, 1.9× decode. FA2 inside FP16 adds 1.07–1.11×. For short-answer VQA, FA2 is not the lever — but it would scale at long context.

Practical insights

1. **For deployment today:** `blip2-flan-t5-xl` or `blip2-opt-2.7b`. The 7B+ tier's quality gain does not justify $\sim$15$\times$ cost.
2. **Optimize the decoder, not the vision tower**, on every model $\geq$ 7B.
3. **Match the optimization to generation length** via N^*: short answers $\rightarrow$ prefill-side (FA2, chunked prefill); long answers $\rightarrow$ decode-side (speculative, INT8/INT4, KV-cache quant).
4. **FP16 first, attention kernels second.** Dtype flip is a 2–4$\times$ lever; FA2 inside FP16 is 1.07–1.11$\times$ at short context.
5. **T5 is a trap with current tooling**: `torch.compile` fails, FP16 makes it slower.
6. **Fuyu's design choice has a cost**: tokenized image patches inflate KV cache $\rightarrow$ 41 ms/token despite being 8B.

Limitations and what's next

Limitations

- FLOPs for T5 backbones estimated as $2 \times$ params; calflops doesn't trace cleanly.
- Energy from `nvidia-smi` at 100 ms polling — a few percent precision, not lab-grade.
- No quantization (INT8/INT4) measured.
- FA2 evaluated only at short context; benefit at $N \gg 1k$ not captured.

Next steps

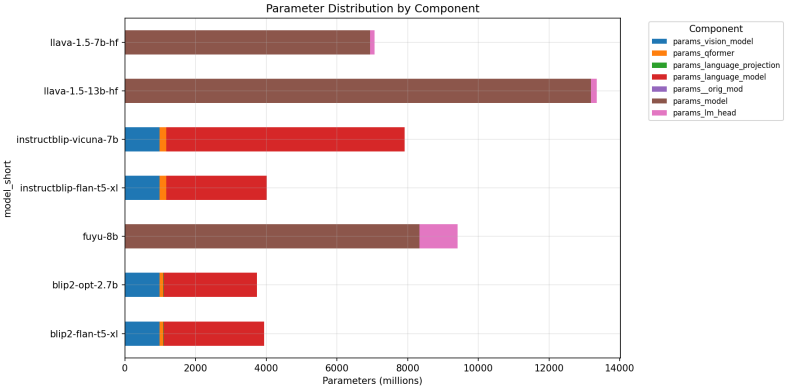
- INT8 / GPTQ / AWQ on 7B+ tier (decode-bound, biggest win).
- Speculative decoding with a small draft model.
- Long-context FA2 sweep (resolution 1024+, multi-image prompts).
- Fix the `idefics2-8b` integration.

Backup: per-dataset WER

Model	COCO Caption	ScienceQA	TextVQA
blip2-flan-t5-xl	0.79	1.30	1.54
blip2-opt-2.7b	0.82	1.49	1.87
instructblip-flan-t5-xl	1.85	1.28	**0.86**
instructblip-vicuna-7b	**0.96**	11.18	6.36
fuyu-8b	4.65	6.42	7.57
llava-1.5-7b-hf	3.37	7.02	8.75
llava-1.5-13b-hf	3.36	7.96	8.06

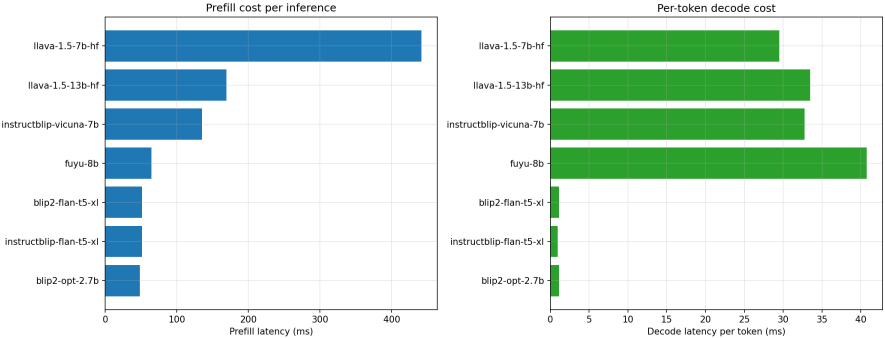
`instructblip-vicuna-7b`: great on COCO (0.96) but breaks on ScienceQA (11.18) — likely an instruction-tuning gap on science-style prompts.

Backup: parameter breakdown



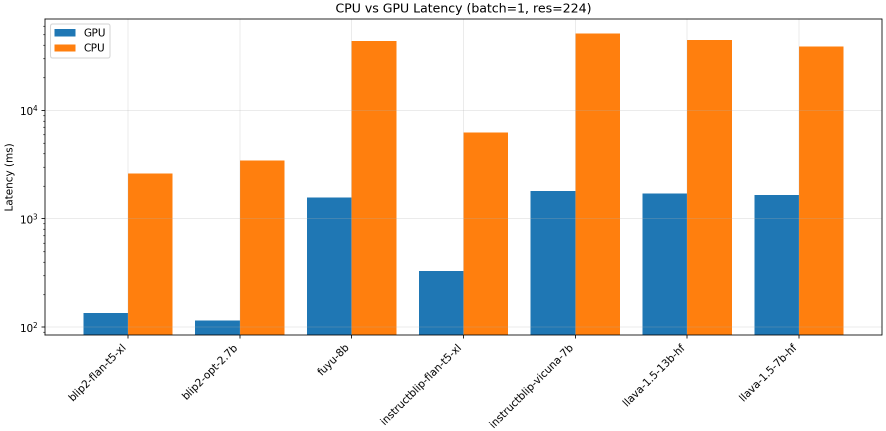
Language decoder dominates parameter count. Vision encoder < 400 M. Matches the runtime breakdown.

Backup: prefill vs. decode (absolute)



`llava-1.5-13b` is in FP16 (does not fit in 48 GiB FP32). Re-running 7B in FP16 confirms the dtype flips the ordering.

Backup: CPU vs GPU



Thanks

Questions?

Repo: `github.com/smirnovlad/vlm-profiler`
Report: `report.pdf`